Technical Report

a. Reason to choose Values for Canny Edge Detector and Hough Transform

- An initial Both-Lanes Detection Rate of 60.0% was observed. After adjusting the **Canny edge detection** thresholds to $\tau_low = 45$ and $\tau_high = 145$, the Both-Lanes Detection Rate improved to 80.0%. The ratio between the high and low thresholds was selected in accordance with the recommendations from Canny's 1986 paper, which suggests using a lower threshold at approximately one-third of the higher threshold. The reduced low threshold ($\tau_low = 45$) was empirically chosen to capture faint lane markings with gradient magnitudes in the 30–50 range, which were missed when using the original threshold of 50. Through hysteresis thresholding, these weak edges are preserved only when connected to strong edges (gradient > 145), effectively suppressing isolated noise while maintaining continuous and valid lane segments.
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b. The **averaging method** was selected for line fitting because it is the Maximum Likelihood Estimator (MLE) under Gaussian noise and achieves minimum variance by the Gauss-Markov theorem. The Kaggle dataset exhibits very low noise ($\sigma \approx 0.0002$) and no outliers, making homoscedastic noise assumptions valid.

   For Gaussian noise: $\varepsilon_i \sim N(0, \sigma^2)$
   Log-likelihood: $\log L = -n/2 \cdot \log(2\pi\sigma^2) - 1/(2\sigma^2) \cdot \Sigma(y_i - \hat{y})^2$
   Maximizing: $d(\log L)/d\theta = 0 \implies \hat{y} = (1/n) \cdot \Sigma(y_i)$
   Conclusion: Averaging is the OPTIMAL estimator

c. System Performance Analysis: The observed output for 30 images from Kaggle data set is as below. But in the ipynb file its run for 20 images just to keep the file siege less than 10MB for submission.

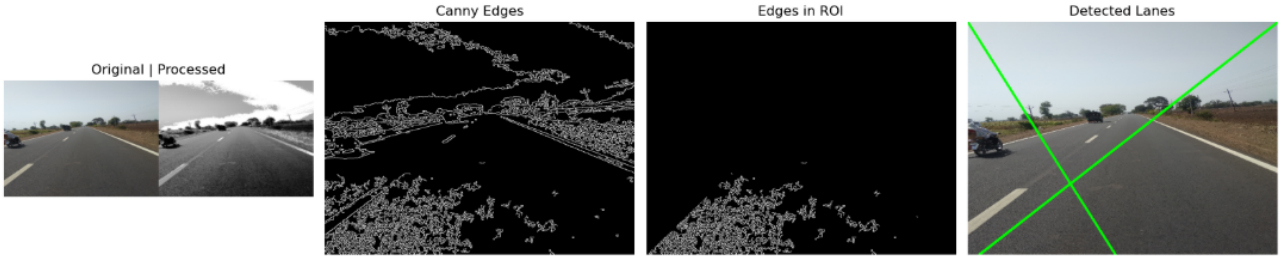
Method: AVERAGING

Total Test Images: 30
Left Lane Detection Rate: 86.7%
Right Lane Detection Rate: 86.7%
Both Lanes Detection Rate: 80.0%
Average Lines per Image: 46.1
✓ Metrics computed

Method: AVERAGING

Total Test Images: 20
Left Lane Detection Rate: 80.0%
Right Lane Detection Rate: 90.0%
Both Lanes Detection Rate: 80.0%
Average Lines per Image: 52.0
✓ Metrics computed

Success case :



Failed Case:
This pipeline is optimised for highway lane detection but it fails when the road has lots of vehicles, barriers or shadows. The image shows where the pipeline has failed to detect the lines

